

tions to determine if the network will work for the desired application.

Measurement devices for optical networks

Smaller and portable microscopes are used to test fibre cables easily. There are two types of microscopes, namely, optical and video. An optical microscope incorporates both an objective and an eyepiece lens to view the end-face directly through the device. A video microscope utilises an optical probe and a display for viewing the probe's image.

A probe can reach hard-to-access places due to its small size. A screen allows images to be expanded for easier identification of contaminants and damage. Because the end-face is viewed on a screen instead of directly, a probe minimises the chance of a harmful laser light reaching a person's eye.

Fibre monitoring helps assess fibre-optic performance metrics over time through the use of software tools and devices that comprise an integrated fibre monitoring and management system. These elements collectively facilitate the detection of faults, attenuation, degradation or security intrusions, and alarm the system administrator in real time when threats to the fibre-optic network integrity occur.

Fibre-optic cables are routinely tested after installation, spliced and terminated to test for continuity and polarity, checked for end-to-end insertion loss and then troubleshooting is done in case of any problems. Testing fibre-optics requires tools and instruments that must be chosen as per the components or cable plants being tested.

Fibre-optic testers like verification testers, certification testers and advanced OTDR testers perform basic inspection, troubleshooting and analysis of an existing fibre-optic cabling. Some fibre-optic cable troubleshooting tools have built-in results storage and automatic wavelength synchronisation that can save time and prevent errors.

Transmitter and receiver power in a network are measured to check whether the system is operating properly or not. Measuring power requires only a power meter, but the meter must be set to the proper range and wavelengths, matching the source being used. Power

measured by the meter must be compared to the specified power for the system to make sure it does not exceed the specified limit.

Oscilloscopes such as Keysight's Infiniium UXR series are used for high-speed serial and optical networking test, measurement and development. These have analogue bandwidth in the range of 13GHz to 110GHz and 10-bit vertical resolution. This ensures signal integrity for the superior effective number of bits, leading to faster compliance testing and measurement.

Optical network monitoring system

An optical network monitoring system accurately detects and locates fibre degradation, alerting operators and managers with the details of faults. Measurement at desired intervals allows network operators to assess long-term fibre performance and efficient asset management. It is designed to be easily integrated with Web-enabled applications, multiple APIs and domains architecture within an organisation.

With increase in data rates, the system becomes more complex and many factors can impair system performance. This makes testing more complicated. For a network that can support high data rate applications as well as extended wavelength range operation for dense wavelength division multiplexing and coarse wavelength division multiplexing systems, it is important



Fig. 1: Automated SYSTIMAX fibre calculator by Fluke Networks

to carry out full fibre characterisation.

Passive optical network (PON) monitoring gives operators the ability to track, test and troubleshoot the entire fibre network. PON monitoring saves substantial time to repair and capital expenses during initial fibre plant construction, network operations, maintenance, fibre-to-the-home (FTTH) network deployments for single-family dwellings and multiple-density units, and any necessary field services.

PON fibre monitoring uses very short pulse

widths of light to receive backscatter and reflection signatures from the network. The shorter the pulse of light used, the higher the detection resolution becomes for probing the network.

Continuous trace monitoring rapidly identifies discrepancies from the reference trace and issues an alarm to dispatch the correct team to resolve the event, delivering maximum customer uptime. Reflectors can also be used to enhance visibility of optical network terminals under low-resolution or high-noise conditions.

The remote fibre monitoring system offered by M2 Optics ensures the performance and integrity of the optical fibre infrastructure. It can remotely monitor up to 16 individual fibre links per device with both dark and lit fibres. It is optimised for point-to-point fibre links up to 160 kilometres. It has integrated optical switching for continuous and/or on-demand link testing with an easy-to-deploy and easy-to-use setup. It can also be scaled up as the network grows.

Optical time domain reflectometer

An OTDR utilises high-intensity laser light emitted at a pre-defined pulse interval through a connecting cable at one end of the fibre-optic cable run. It analyses the backscattered light returning to the source location. It is possible to take measurements over relatively

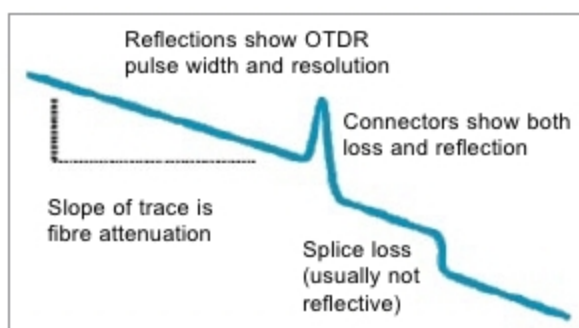


Fig. 2: OTDR display (Credit: www.lanshack.com)